MSG No. 183.190 P_c6mm [Type IV, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `6mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `4b`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,3,5,7,8,9]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[2,4,6,10,11,12]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[13,15,17,19,20,21]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[14,16,18,22,23,24]`

Table 3: Wyckoff site: `6c`, site symmetry: `2mm`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4,7,11]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[2,5,9,10]`
3	$\left[0,\ \frac{1}{2},\ z\right]$	`[3,6,8,12]`
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[13,16,19,23]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[14,17,21,22]`
6	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,18,20,24]`

Table 4: Wyckoff site: `12d`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,11]`
2	$[x,\ x,\ z]$	`[2,9]`
3	$[0,\ x,\ z]$	`[3,12]`
4	$[-x,\ 0,\ z]$	`[4,7]`
5	$[-x,\ -x,\ z]$	`[5,10]`
6	$[0,\ -x,\ z]$	`[6,8]`
7	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	`[13,23]`
8	$\left[x,\ x,\ z+\frac{1}{2}\right]$	`[14,21]`
9	$\left[0,\ x,\ z+\frac{1}{2}\right]$	`[15,24]`
10	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[16,19]`
11	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	`[17,22]`
12	$\left[0,\ -x,\ z+\frac{1}{2}\right]$	`[18,20]`

Table 5: Wyckoff site: `12e`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ -x,\ z]$	`[1,8]`
2	$[2x,\ x,\ z]$	`[2,11]`
3	$[x,\ 2x,\ z]$	`[3,9]`
4	$[-x,\ x,\ z]$	`[4,12]`
5	$[-2x,\ -x,\ z]$	`[5,7]`
6	$[-x,\ -2x,\ z]$	`[6,10]`
7	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	`[13,20]`
8	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	`[14,23]`
9	$\left[x,\ 2x,\ z+\frac{1}{2}\right]$	`[15,21]`
10	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	`[16,24]`
11	$\left[-2x,\ -x,\ z+\frac{1}{2}\right]$	`[17,19]`
12	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	`[18,22]`

Table 6: Wyckoff site: `24f`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x-y,\ x,\ z]$	`[2]`
3	$[-y,\ x-y,\ z]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$[-x+y,\ -x,\ z]$	`[5]`
6	$[y,\ -x+y,\ z]$	`[6]`
7	$[-x+y,\ y,\ z]$	`[7]`
8	$[-y,\ -x,\ z]$	`[8]`
9	$[x,\ x-y,\ z]$	`[9]`
10	$[-x,\ -x+y,\ z]$	`[10]`
11	$[x-y,\ -y,\ z]$	`[11]`
12	$[y,\ x,\ z]$	`[12]`
13	$\left[x,\ y,\ z+\frac{1}{2}\right]$	`[13]`
14	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	`[14]`
15	$\left[-y,\ x-y,\ z+\frac{1}{2}\right]$	`[15]`
16	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[16]`
17	$\left[-x+y,\ -x,\ z+\frac{1}{2}\right]$	`[17]`
18	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	`[18]`
19	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	`[19]`
20	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[20]`
21	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	`[21]`
22	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	`[22]`
23	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	`[23]`
24	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[24]`